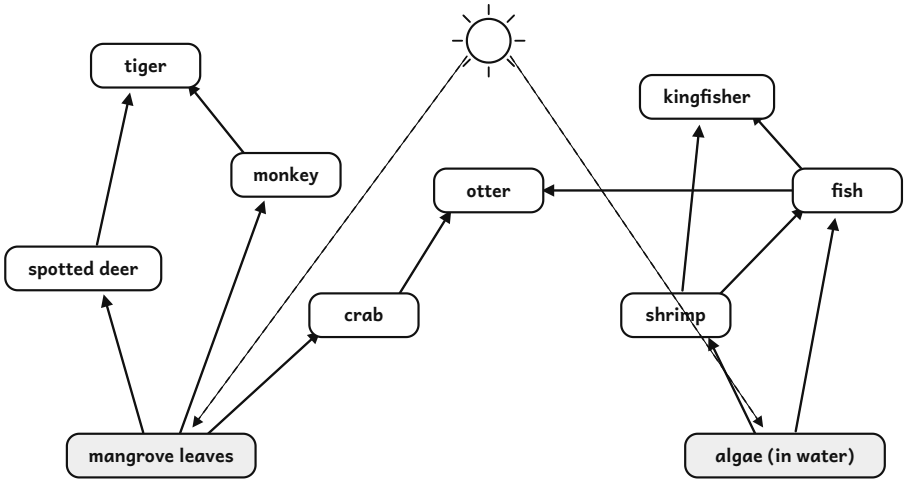
LIVING THINGS

Living Things in Their Environment

Habitats, biodiversity, food webs, human impact and learning from nature



Arrows show the direction energy moves: from the food to the animal that eats it.

In this pack

- **Learn it:** Habitats, food webs and our place in nature ★☆☆
- **Fact check:** Habitats and food webs ★☆☆
- **Think it through:** What happens if…? ★★☆
- **Investigate:** Comparing two habitats ★★☆
- **Real-world case study:** The Sundarbans: a forest that protects us ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Habitats, food webs and our place in nature

★☆☆ Grade 5

Name: Class: Date:

Learning goals

- Compare the living things found in different habitats.
- Use a food web to explain how energy and matter move through an ecosystem.
- Evaluate how human activities affect ecosystems, and how nature inspires design.

A pond, a mango orchard, a tea garden in Sylhet, the coral reef at St Martin's Island and the mangrove forest of the Sundarbans are all **habitats**: places where living things find food, water and shelter. All the living things in a habitat, together with the non-living things around them (water, soil, light, air), make an **ecosystem**.

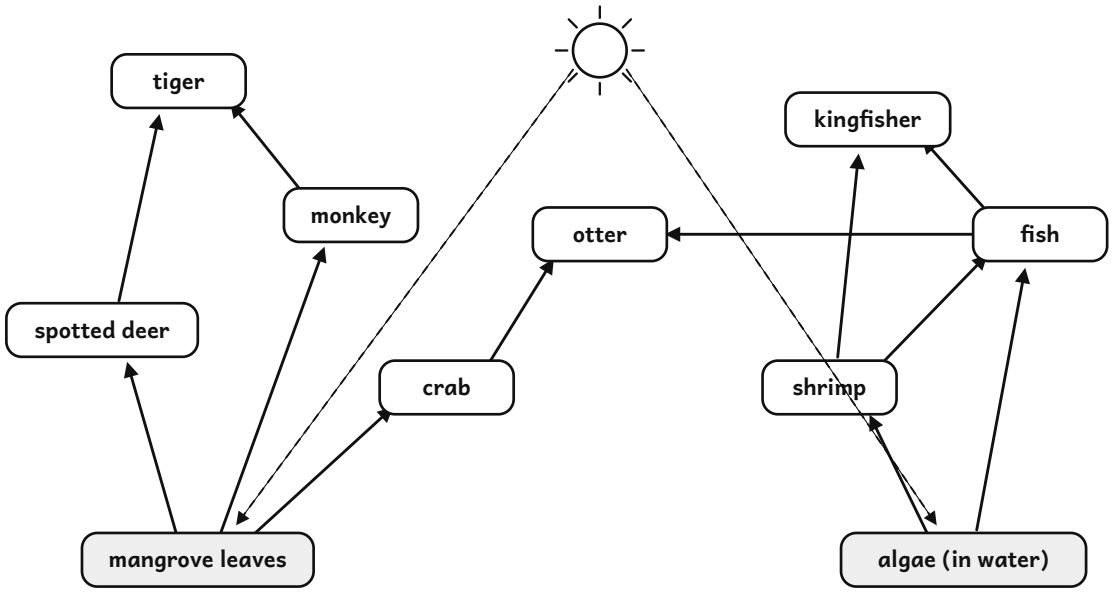
Biodiversity

Biodiversity means the variety of living things in a place. A rainforest or a coral reef has huge biodiversity. A field with only one crop has very little. Ecosystems with high biodiversity are usually healthier and recover better from problems such as disease or floods.

Food chains and food webs

Every food chain starts with a **producer**, a green plant or alga that makes its own food using light energy from the Sun. **Consumers** eat other living things: **herbivores** eat plants, **carnivores** eat animals and **omnivores** eat both. A carnivore that is not eaten by anything else is a **top predator**.

In a real habitat, most animals eat more than one kind of food, so food chains join together into a **food web**. The arrows show the direction in which **energy** moves: from the food to the animal that eats it.



Arrows show the direction energy moves: from the food to the animal that eats it.

A simplified food web in the Sundarbans.

Energy flows, matter cycles

Energy enters the ecosystem as sunlight and passes along the food chains. At each step, much of the energy is used for living (moving, keeping warm), so there is less for the next animal. That is why there are fewer tigers than deer. **Matter** is different: it goes round and round. When plants and animals die, **decomposers** such as fungi, bacteria and earthworms break them down, returning nutrients to the soil for new plants.

Living things depend on each other

Bees pollinate flowers; birds spread seeds; trees give shelter to monkeys and birds; crabs clean up dead leaves. When one species disappears, others are affected. Reproduction is important too: if a species cannot breed, its numbers fall and the whole web changes.

Humans and ecosystems

People change habitats by cutting forests, building, farming, fishing and polluting rivers and seas with chemicals and plastic. We can also protect them: national parks, tree planting, fishing rules and cleaner factories all help.

Learning from nature

Engineers often copy nature. This is called **biomimicry**. The hooks on burrs inspired **Velcro**. The shape of the kingfisher's beak inspired the quiet nose of Japan's bullet train. Termite mounds inspired buildings that stay cool without air conditioning. Lotus leaves inspired self-cleaning paints.

Bangladesh connection

Bangladesh has more than 700 species of birds, fish and mammals living in its rivers, haors, hills and mangroves. The Sundarbans is a UNESCO World Heritage Site and home to the Royal Bengal tiger.

Key words

habitat	the place where a living thing lives
ecosystem	living things and their surroundings working together
biodiversity	the variety of living things in a place
producer	a plant or alga that makes its own food
consumer	an animal that eats other living things
decomposer	a living thing that breaks down dead material
food web	food chains linked together
biomimicry	copying nature to solve problems

FACT CHECK · A7 · ENVIRONMENTS

Habitats and food webs

★☆☆ Grade 5

Name: Class: Date:

Learning goals

- Use the key vocabulary.
- Read a food web.

1 Match each word to its meaning. [6]

Draw a line to match each pair, or write the letter.

producer ●	● **A** eats only animals
herbivore ●	● **B** eats plants and animals
carnivore ●	● **C** the place where an organism lives
omnivore ●	● **D** breaks down dead plants and animals
decomposer ●	● **E** eats only plants
habitat ●	● **F** makes its own food using light

2 Use the food web on the "Learn it" page to answer. [4]

a) Name the two producers.

.....

b) Write one food chain with four links.

.....

c) Name a top predator.

.....

d) What do crabs eat?

.....

3 **True or false?** [4]

Statement	True	False	If false, correct it
1. Every food chain starts with a producer.			
2. The arrows in a food web point from the eater to the food.			
3. Decomposers return nutrients to the soil.			
4. A field with one crop has high biodiversity.			

4 **Fill in the gaps.** [4]

WORD BANK

biomimicry **sunlight** **consumers** **soil** **food** **rain**

Energy enters a food web as _____. Plants use it to make _____. Energy passes to _____ when they eat. Copying ideas from nature is called _____.

What happens if...?

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Predict changes in a food web.
- Explain energy flow.
- Link features of living things to designs.

1 **Use the Sundarbans food web. Predict what would happen and explain why.** [4]

a) If many deer died from disease, what would happen to the tigers?

b) If an oil spill killed the algae, which animals would be affected? Explain.

2 **Why are there far fewer tigers than deer in the Sundarbans?** [2]

3 **Complete the table about biomimicry.** [6]

Feature in nature	What it does	Human invention
burr hooks		
kingfisher beak		
termite mound		

4 **Design a solution by mimicking nature. A school roof gets very hot. Which animal or plant could give you an idea? Explain your design.** [2]

INVESTIGATE · A7 · ENVIRONMENTS

Comparing two habitats

★★☆ Grade 5

Name: Class: Date:

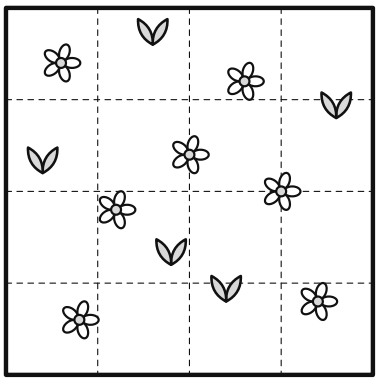
Learning goals

- Use a quadrat to sample plants fairly.
- Compare biodiversity using data.
- Explain differences between habitats.

Class 5A compared two habitats in the school grounds: the **football field** and the **garden by the pond**. In each habitat they threw a 1 m × 1 m **quadrat** at random three times and counted the different kinds of plants and small animals inside.

Remember

Throwing the quadrat at random makes the test fair: you do not choose only the interesting spots.

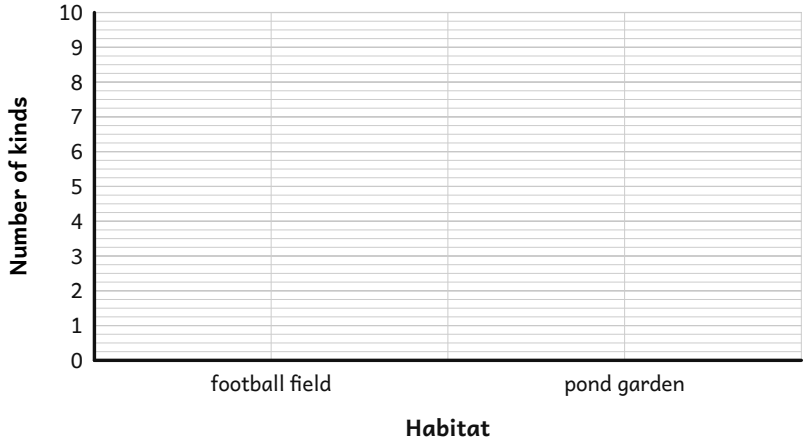


1 metre × 1 metre quadrat

1 **Here are the results. Work out the totals.** [2]

Kind of living thing	Football field	Pond garden
grass	✔	✔
clover	✔	✔
dandelion		✔
fern		✔
moss		✔
ant	✔	✔
earthworm		✔
snail		✔
spider		✔
Number of different kinds		

2 **Draw a bar chart comparing the number of different kinds of living things.** [3]



3 **Which habitat has greater biodiversity? Suggest two reasons.** [3]

.....

.....

.....

4 **Why did the class use the quadrat three times in each habitat?** [1]

.....

.....

5 **Suggest one way the school could increase biodiversity on the field.** [1]

.....

.....

REAL-WORLD CASE STUDY · A7 · ENVIRONMENTS

The Sundarbans: a forest that protects us

★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Explain why an ecosystem matters to people.
- Evaluate the effect of a human activity, using evidence.
- Argue for a way to reduce harm.

Science · Geography · Economics · Ethics · Climate change

The **Sundarbans** is the largest mangrove forest in the world, about 10 000 km², shared by Bangladesh (about 60%) and India. Mangrove trees such as *sundari* and *keora* grow in salty, muddy water and have special roots that poke up out of the mud to breathe.

The forest is home to Royal Bengal tigers, spotted deer, crocodiles, river dolphins and hundreds of kinds of birds and fish. Hundreds of thousands of people depend on it for fish, crabs, honey and wood.

The Sundarbans also **protects** people. When cyclones such as **Sidr** (2007) and **Aila** (2009) struck, the mangroves slowed the wind and the storm surge, protecting villages behind them.

But the forest faces threats. In **December 2014**, an oil tanker sank in the Shela River inside the forest, spilling about 350 000 litres of heavy oil. Oil coated mud, roots and animals. Rising sea levels, more salt in the water, shipping, cutting of trees and poaching are other dangers.

Sources: UNESCO World Heritage listing; UNEP/UN assessment of the 2014 Shela River oil spill.

1 **Give three ways the Sundarbans helps people.** [3]

2 **Explain how an oil spill could harm the food web, starting with the producers.** [3]

3 **Different people want different things from the river. Complete the table.** [3]

Person	What they want
a ship owner	
a fisher	
a conservation scientist	

4 **Argue from evidence. Should cargo ships be allowed to travel through the Sundarbans? Give your opinion, one piece of evidence and one way to reduce harm.** [4]

Show what you know

★★★ Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

M	E	V	P	R	O	D	U	C	E	R	I
D	L	E	N	E	R	G	Y	U	M	L	P
D	E	D	P	O	P	P	J	C	A	E	D
K	I	C	P	Y	F	R	A	G	N	R	L
E	Y	C	O	S	A	R	J	V	G	C	Y
I	R	L	O	M	H	F	L	H	R	S	S
R	K	V	H	N	P	A	U	G	O	H	M
H	T	G	R	P	S	O	B	L	V	A	A
I	I	P	I	G	Y	U	S	I	E	U	L
O	G	L	P	R	E	Y	M	E	T	L	C
H	E	D	H	P	G	K	G	E	R	A	P
U	R	P	R	E	D	A	T	O	R	U	T

consumer **decomposer** **energy** **habitat** **mangrove** **predator** **prey** **producer** **tiger**

2 **In the chain grass → deer → tiger, the deer is:** [1]

- ◯ a producer
- ◯ a herbivore
- ◯ a carnivore
- ◯ a decomposer

3 **Which is an example of biomimicry?** [1]

- ◯ planting trees
- ◯ Velcro copied from burrs
- ◯ a food chain
- ◯ a national park

4 **Draw a food web with at least five living things from a pond. Use arrows correctly.** [4]

My pond food web

5 **Challenge. Explain why matter is recycled in an ecosystem but energy is not.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
compare the variety of living things in two habitats	○	○	○
read and draw a food web with correct arrows	○	○	○
explain how energy flows and matter cycles in an ecosystem	○	○	○
predict what happens when one living thing disappears	○	○	○
explain how a human activity harms an ecosystem and argue for a solution	○	○	○
give examples of biomimicry	○	○	○